Naghmeh (Melody) Nazar

FALL 2027 PhD APPLICANT | AI SAFETY & TOOL-USING AGENT SECURITY | SPEECH ML

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RESEARCH FOCUS

Fall 2027 PhD applicant studying pre-execution safeguards for indirect prompt injection from untrusted email, documents, and tool outputs in tool-using large language model (LLM) agents. Current independent work maps selected MCP-SafetyBench attack paths involving untrusted tool metadata, user input, and tool outputs and is designing an evaluation using attack success, benign task completion, false-block rate, and calibration. M.S. thesis: four-class speech emotion recognition on RAVDESS and TESS, with explicit limits on speaker-independent and cross-corpus claims. Brings 3+ years of production software engineering experience to building testable AI systems.

Research interests: LLM agent security, AI safety, indirect prompt injection, scalable oversight, adversarial evaluation, speech processing, affective computing

RESEARCH EXPERIENCE

Independent Research - Pre-Execution Safeguards for Tool-Using LLM Agents **Current**

- Mapping selected MCP-SafetyBench attack paths (Zong et al., ICLR 2026) involving untrusted tool metadata, user input, and tool outputs as they propagate to proposed tool selection and arguments. MCP-SafetyBench paper
- Designing an evaluation of a candidate pre-execution risk-gating baseline; planned measures include attack success, benign task completion, false-block rate, and calibration. Analysis and experiment design are in progress; no validated defense results are claimed.
- Built an n8n email-and-scheduling agent and documented where untrusted messages and generated actions require validation or human approval. github.com/nzrnaghme/RespondEmailAgent

M.S. Thesis - Speech Emotion Recognition, CSUN **Aug 2026**

- Developed a four-class 2D ResNet-inspired CNN on RAVDESS and TESS using 16 kHz audio and 128 x 128 log-Mel spectrograms.
- On the current mixed-corpus split, achieved 80.36% validation accuracy and 74.4% held-out test accuracy; the split does not establish speaker-independent or cross-corpus generalization.
- Desktop latency and naturalistic-speech generalization remain unmeasured; no deployment or clinical-validation claim is made.

RESEARCH OUTPUTS & PRESENTATION

- M.S. thesis - "Real-Time Speech Emotion Recognition Using Optimized Deep Residual Networks," California State University, Northridge, Aug 2026. PDF available on request.
- Presenter - 40th Annual CSU Student Research Competition: M.S. speech emotion recognition research, 2026. Presentation materials available on request.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge **Aug 2024 - Aug 2026**

Thesis advisor: Prof. Shahnam Mirzaei

B.S. in Computer Engineering, University of Science and Technology of Mazandaran **Aug 2016 - Mar 2021**

PROFESSIONAL EXPERIENCE

Senior Frontend Developer, Golrang Industrial Group - Tehran, Iran **Jul 2022 - Jul 2024**

- Architected cross-platform applications with React Native and React.js PWA; contributed UI implementation, state management, caching, performance optimization, and code review.

React Developer, Erole.ir - Tehran, Iran **Jun 2021 - Jun 2022**

- Built React and Redux applications and integrated REST APIs and SignalR-based real-time features.

TECHNICAL SKILLS

Agent-security study: indirect prompt injection, attack-path analysis, tool-call risk evaluation design

Speech and machine learning: Python, TensorFlow, Keras, scikit-learn, NumPy, Pandas, 2D CNN/ResNet, log-Mel spectrograms, MFCCs

Engineering: C++, JavaScript, TypeScript, n8n, React, React Native, Node.js, REST APIs, Git, Linux